

DEBMALYA SUR

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[Website](#) ◇ [LinkedIn](#) ◇ [GitHub](#) ◇ [Leetcode](#)

EDUCATION

Indian Institute of Technology, Dhanbad

Master of Technology in Computer Science and Engineering

CGPA: 9.26/10

Government College of Engineering & Ceramic Technology, Kolkata

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.87/10

PROFESSIONAL EXPERIENCE

Senior Engineer — Samsung Research Institute Bangalore (SRIB)

Jul 2024 – Present

- Architecting and leading KV-cache orchestration for the PersonalAgentCore (PAC) framework, achieving a 97%+ reduction in context-switch latency, from 3.2 seconds to < 100 ms.
- Owned the end-to-end on-device machine learning pipeline for the SuggestStyle feature, commercially launched on Samsung Galaxy S24 Dress Room, including model development, optimization, deployment, and production integration.
- Developed a lightweight anchor-free object detection solution using NanoDet and an aesthetics assessment model, achieving 20ms end-to-end on-device inference latency for real-time user experiences.
- Optimized the production model from 8MB to 2MB through model compression and ONNX-based optimization techniques while maintaining minimal performance degradation, enabling deployment on resource-constrained mobile devices.
- Designed and implemented a personalized prompt generation pipeline leveraging user preference data (TPO and PDE signals) from Samsung Account services, generating context-aware prompts for AI wallpaper recommendations.
- Developing a server-side vision-language classification system in collaboration with the Samsung Ads team, utilizing CLIP-based encoders to automatically categorize advertisement content into SFW and NSFW classes with 95% accuracy, less than 2% FPR and less than 5% FNR and reduce the manual intervention by 90%

Research Intern — Samsung Research Institute Bangalore (SRIB)

Jun 2023 – Aug 2023

- Improved render-thread performance and reduced UI latency by approximately 15% for on-device AI applications through graphics optimization techniques.
- Designed and implemented a Huffman Encoding-based texture optimization algorithm achieving up to 93% compression while preserving visual quality.

PROJECTS

Baud News

- Designed and developed a full-stack news aggregation platform using Flask, Redis, Firebase, and SQLite, enabling personalized content discovery through recommendation-driven news feeds.
- Architected automated content ingestion and ranking pipelines for collecting, processing, and recommending news articles based on user interests and content relevance.

SKILLS

Core Areas: Computer Vision, Vision-Language Models (CLIP), Object Detection, On-Device AI, Model Compression & Optimization, Agentic AI Systems, Large Language Models (LLMs), Machine Learning, Deep Learning, MLOps

Frameworks & Tools: PyTorch, OpenCV, Flask, Redis, Firebase, MongoDB, SQLite, Git, Ollama

Programming Languages: Python, C++, Java

ACHIEVEMENTS & PUBLICATIONS

- Received the Feature Commercialization Award for Samsung Galaxy S24 for model-side end-to-end ownership and successful commercialization of the SuggestStyle feature.
- Authored and published research on “A Clustered Federated Learning Approach for Heart Failure Prediction” on [SSRN](#), proposing privacy-preserving machine learning techniques for distributed healthcare data analytics.
- Authored and published peer-reviewed research on “Blockchain-based Authentication and Key Agreement Scheme for Securing IoMT Device Network” in [Elsevier](#)’s Blockchain: Research and Applications journal, enhancing device security and secure communication.
- Secured All India Rank (AIR) 501 in GATE CS 2022.